
Viscous-Hydrodynamics from Decoherence

Caldeira-Leggett Master Equation

- Configuration: Single system in presence of Environment at finite Temperature T (The Environment is modeled as infinite set of Harmonic oscillators linearly coupled, The oscillators represents environments degree of freedom, Also we Assume the system is in potential $V(x)$) .

- $$H_{total} = \frac{p^2}{2m} + V(\bar{x}) + \sum_n \left(\frac{p_n^2}{2m_n} + \frac{1}{2} m_n \omega_n^2 x_n^2\right) - \bar{x} \sum_n c_n \bar{x}_n + x^2 \sum_n \frac{c_x^2}{2m_n \omega_n^2}$$

- $H_{total} = H_S + H_E + H_{int} + H_c$

- H_c is the counter-term which leads to renormalization of the systems potential. This counter term is introduced to prevent un-physical divergence.

- Applying this Hamiltonian to Von-Neumann time evolution equation i.e $\frac{d}{dt}\rho_{total}(t) = -\frac{i}{\hbar}\left[H_{total}(t), \rho_{total}(t)\right]$ gives us the master equation .

- Caldeira-Leggett master equation [35]

$$\frac{\partial}{\partial t}\rho_S(t) = -\frac{i}{\hbar}[H_S, \rho_S(t)] - \frac{i\gamma}{\hbar}[x, \{p, \rho_S(t)\}] - \frac{2m\gamma k_B T}{\hbar^2}[x, [x, \rho_S(t)]].$$

- The Caldeira-Leggett master equation in the coordinate representation is given by

$$\frac{\partial}{\partial t}\rho_S(\vec{x},\vec{x}\,',t) = \frac{i\hbar}{2m}\left(\nabla^2 - \nabla'^2\right)\rho_S(\vec{x},\vec{x}\,',t) - \frac{i}{\hbar}(V(\vec{x}) - V(\vec{x}\,'))\rho_S(\vec{x},\vec{x}\,',t) - \gamma(\vec{x}-\vec{x}\,')(\nabla - \nabla')\rho_S(\vec{x},\vec{x}\,',t) - \frac{2m\gamma k_B T}{\hbar^2}(\vec{x}-\vec{x}\,')^2\rho_S(\vec{x},\vec{x}\,',t),$$

- where $\rho_S(\bar{x},\bar{x}',t) = \langle \bar{x} | \rho(t) | \bar{x}' \rangle$

$$\frac{\partial}{\partial t}\rho_S(\vec{x},\vec{x}\,',t) = \frac{i\hbar}{2m}\left(\nabla^2 - \nabla'^2\right)\rho_S(\vec{x},\vec{x}\,',t) - \frac{i}{\hbar}(V(\vec{x}) - V(\vec{x}\,'))\rho_S(\vec{x},\vec{x}\,',t)$$

$$- \boxed{\gamma(\vec{x}-\vec{x}\,')\left(\nabla - \nabla'\right)\rho_S(\vec{x},\vec{x}\,',t)} - \boxed{\frac{2m\gamma k_B T}{\hbar^2}(\vec{x}-\vec{x}\,')^2\rho_S(\vec{x},\vec{x}\,',t),} \quad (21)$$

Damping Term
models dissipative Energy to E.

Diffusion term
represent thermal fluctuation from E.
Causing the quantum state to spread by Decohere.

Decoherence

- At High enough Temperature of the Environment the 4th term in the previous equation D=dominates. Resulting
- $\frac{d}{dt}\rho_s(\bar{x}, \bar{x}', t) \approx -\gamma\left(\frac{(\Delta x)}{\lambda_{th}}\right)^2 \rho_S(\bar{x}, \bar{x}', t) \quad where\ \lambda = \frac{\hbar}{\sqrt{2mK_BT}}$
- i.e $\rho_S(\bar{x}, \bar{x}', t) \approx e^{\Gamma(t)}\rho_S(\bar{x}, \bar{x}', 0)$ for $\Delta x \gg \lambda_{th}$ (large distance form diagonal)

Hydro and Caldeira-Leggett ME

- Let $\rho_S = e^{R+iS}$ *while making sure* $\rho_S^\dagger = \rho_S$
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$$\partial_t R = -\frac{\hbar}{2m}\left(\nabla_+\cdot\nabla_- S + \nabla_+ R\cdot\nabla_- S + \nabla_+ S\cdot\nabla_- R\right) - 2\gamma\vec{x}_-\cdot\nabla_- R - \frac{8m\gamma k_B T}{\hbar^2}\vec{x}_-^{\,2},$$

$$\partial_t S = \frac{\hbar}{2m}\left(\nabla_+\cdot\nabla_- R + \nabla_+ R\cdot\nabla_- R - \nabla_+ S\cdot\nabla_- S\right) - \frac{1}{\hbar}\left(V(\vec{x}_+ + \vec{x}_-) - V(\vec{x}_+ - \vec{x}_-)\right) - 2\gamma\,\vec{x}_-\cdot\nabla_- S,$$

center-of-mass and relative coordinates

$$\vec{x}_+ = \frac{\vec{x}+\vec{x}\,'}{2}, \quad \vec{x}_- = \frac{\vec{x}-\vec{x}\,'}{2},$$

$$\nabla_- = \nabla - \nabla', \quad \nabla_+ = \nabla + \nabla'.$$

Taylor-Expansion of ρ_S

- Decoherence suggested the off-diagonal terms decay exponentially thus allowing us to Taylor expand $\rho_{\bar{x},\bar{x}',t}$ around point $\bar{x} = \bar{x}'$.
- $$\rho_S(\vec{x}, \vec{x}\,', t) = \rho_S(\vec{x}, \vec{x}, t) + \sum_i \left.\frac{\partial \rho_S}{\partial x_i}\right|_{\vec{x}=\vec{x}\,'} (x_i - x_i') + \frac{1}{2} \sum_{i,j} \left.\frac{\partial^2 \rho_S}{\partial x_i \partial x_j}\right|_{\vec{x}=\vec{x}\,'} (x_i - x_i')(x_j - x_j') + \cdots .$$
- $$S = a_i x_-^i + B_{ijk} x_-^i x_-^j x_-^k + \cdots = S^{(1)} + S^{(3)} + \cdots,$$
  $$R = \log n + C_{ij} x_-^i x_-^j + E_{ijkl} x_-^i x_-^j x_-^k x_-^l + \cdots = R^{(0)} + R^{(2)} + R^{(4)} + \cdots,$$

Zeroth-Order term and Local Conservation of Probability

* 0-th order in this equation $\partial_t R = -\frac{\hbar}{2m}\left(\nabla_+ \cdot \nabla_- S + \nabla_+ R \cdot \nabla_- S + \nabla_+ S \cdot \nabla_- R\right) - 2\gamma \vec{x}_- \cdot \nabla_- R - \frac{8m\gamma k_B T}{\hbar^2}\vec{x}_-^{\,2},$ gives:

$$\partial_t \log n = -\frac{\hbar}{2m}\left(\nabla_+ \cdot \vec{a} + \vec{a} \cdot \nabla_+ \log n\right).$$

One may rewrite this equation as a conservation law

$$\partial_t n + \nabla_+ \cdot (n\vec{u}) = 0,$$

where the velocity is

$$\vec{u} = \frac{\hbar}{2m}\vec{a}.$$

$n :=$ *Probablity of state* $\rho_S(x, x)$ $\qquad$ $u :=$ *velocity* $\hbar \nabla S = m\bar{u}$

This is Continuity equation, implying local conservation of probability

First-Order and Momentum Eq

* 1-th order in this equation $\partial_t S = \frac{\hbar}{2m}\left(\nabla_+ \cdot \nabla_- R + \nabla_+ R \cdot \nabla_- R - \nabla_+ S \cdot \nabla_- S\right) - \frac{1}{\hbar}\left(V(\vec{x}_+ + \vec{x}_-) - V(\vec{x}_+ - \vec{x}_-)\right) - 2\gamma\, \vec{x}_- \cdot \nabla_- S,$ gives:

Next, we consider terms of first order in $\mathcal{O}\left(x_-\right)$. By isolating these terms in Eq. (27), we find

$$\partial_t \vec{u} + \left(\vec{u} \cdot \nabla_+\right) \vec{u} = \frac{1}{mn} \nabla_+ \cdot \mathcal{C} - \frac{1}{m} \nabla_+ V - 2\gamma \vec{u}, \tag{36}$$

where we have also used Eq. (35). We can write the above equation in a way that resembles the Euler equation

$$\partial_t (mn\vec{u}) + \nabla_+ \cdot \left(mn\vec{u} \otimes \vec{u} - \mathcal{C}\right) = -n\nabla_+ V - 2\gamma mn\vec{u}, \tag{37}$$

where we have defined

$$\mathcal{C}_{ij} \equiv \frac{n\hbar^2}{2m} C_{ij}. \tag{38}$$

This is Euler equation in build mechanics describing flow/velocity of the fluid

Euler equation/Cauchy momentum equation

- Thus suggest that $C_{i,j}$ is a Stress-Tensor called Cauchy Stress Tensor.

$$\frac{D\mathbf{u}}{Dt} = \frac{1}{\rho}\nabla\cdot\boldsymbol{\sigma} + \mathbf{a}$$

where

- $\mathbf{u}$ is the flow velocity vector field, which depends on time and space, (unit: $\mathrm{m/s}$)
- t is time, (unit: s)
- $\frac{D\mathbf{u}}{Dt}$ is the material derivative of $\mathbf{u}$, equal to $\partial_t\mathbf{u} + \mathbf{u}\cdot\nabla\mathbf{u}$, (unit: $\mathrm{m/s^2}$)
- ρ is the density at a given point of the continuum (for which the continuity equation holds), (unit: $\mathrm{kg/m^3}$)
- $\boldsymbol{\sigma}$ is the stress tensor, (unit: $\mathrm{Pa} = \mathrm{N/m^2} = \mathrm{kg}\cdot\mathrm{m^{-1}}\cdot\mathrm{s^{-2}}$)
- $\mathbf{a} = \begin{bmatrix} f_x \\ f_y \\ f_z \end{bmatrix}$ is a vector containing all of the body accelerations (sometimes simply gravitational

Second-Order and Muller-Israle-Steward Relation:

* 2-th order in this equation $\partial_t R = -\frac{\hbar}{2m}\left(\nabla_+ \cdot \nabla_- S + \nabla_+ R \cdot \nabla_- S + \nabla_+ S \cdot \nabla_- R\right) - 2\gamma \vec{x}_- \cdot \nabla_- R - \frac{8m\gamma k_B T}{\hbar^2}\vec{x}_-^{\,2},$ gives:

$$\partial_t C_{ij} = -\left(\vec{u}\cdot\nabla_+\right) C_{ij} - \left(\nabla_+^k u_i\right) C_{kj} - C_{ik}\left(\nabla_+^k u_j\right) - 4\gamma C_{ij} - \frac{8m\gamma k_B T}{\hbar^2}\delta_{ij},$$

Now the goal is to solve C, at equilibrium $\frac{\partial C_{i,j}}{\partial t} = 0$ and $\bar{u}$ terms vanish if there is n

bulk flow. Thus

$$\mathcal{C}_{ij}^{\mathrm{eq}} = -n\, k_B T \delta_{ij} \equiv -P\delta_{ij}$$

Now do a small perturbation such that : $\mathcal{C}_{ij} = \mathcal{C}_{ij}^{\mathrm{eq}} + \pi_{ij} = -P\delta_{ij} + \pi_{ij},$

$$\tau_\pi \left[\frac{\mathrm{D}}{\mathrm{D}t}\pi_{ij} + \left(\nabla_+^k u_k\right)\pi_{ij} + \left(\nabla_+^k u_i\right)\pi_{kj} + \left(\nabla_+^k u_j\right)\pi_{ik}\right] + \pi_{ij} = 2\eta\sigma_{ij} + \zeta\delta_{ij}\nabla_+^k u_k$$

Where $\dfrac{D}{Dt} = \dfrac{\partial}{\partial t} + \bar{u}\,.\,\nabla_+$

where we defined the relaxation time τ_π, the shear viscosity η, the bulk viscosity ζ, and the shear tensor σ

$$\tau_\pi = \frac{1}{4\gamma}, \quad \eta = \frac{nk_BT}{2}\tau_\pi, \quad \zeta = \frac{4\eta}{3}, \quad \sigma^{ij} = \left(\nabla_+^i u^j + \nabla_+^j u^i - \frac{2}{3}\delta^{ij}\nabla_+^k u_k\right). \tag{47}$$

This eq is the non-relativistic version of Muller-Israle-Steward Relation

Which is used to model hydrodynamics in Heavy-Ion Collion.